Smart Agricultural Rover

A Mobile IoT Controlled Multi-Functional Agricultural Rover

Department of CSE
Course: EEE2124 | Section: C

Instructor Jerin Hasan
Priya (Lecturer)

Problem Statement

Labor Intensive Farming

Traditional agricultural tasks like ploughing, seed sowing, and liquid spraying require separate manual operations, leading to excessive physical strain and time loss.

Inefficient Monitoring

Manual field observation makes early crop hazard detection difficult. Operating heavy equipment manually across large farmlands poses operational challenges.

Proposed Solution

- **Multi-Functional Platform:** Single mobile IoT rover performing soil preparation, seeding, and spraying.
- **Integrated Plough & Seeder:** Mini plough combined with automated servo dispenser.
- **Dual Liquid Delivery:** Switchable water irrigation and pesticide spraying mechanism.
- **IoT Telemetry:** Wireless mobile control and real-time camera feed via ESP32-CAM.



Project Objectives

- **Rover Architecture:** Fabricate a compact, durable mobile chassis for field traversal.
- **Mechanized Ploughing:** Integrate a mini soil-tilling plough mechanism.
- **Precision Seeding & Sensing:** Implement servo dispenser with active seed level indicator.
- **Fluid Spraying & Level Control:** Pump mechanism for water/pesticide with liquid level monitoring.
- **Remote IoT Command:** Live ESP32-CAM video streaming and directional mobile app control.

System Operational Workflow

1

Ploughing

Rover prepares soil with attached mini plough.

2

Seed Sowing

Servo dispenses seeds; level sensor checks container.

3

Watering

Pump sprays water; tank level sensor monitors fluid.

4

Plant Care

Pesticide spraying during crop growth stage.

Continuous Overlay: ESP32-CAM Live Monitoring + Mobile Phone IoT Control

System Architecture

Controller & Telemetry

Mobile App (Wi-Fi) → **ESP32-CAM** (Live Feed)

ESP32 Main Controller: Processes control commands and reads sensors.

Sensors: Seed Level Sensor, Water Tank Level Sensor.

Actuators & Execution

TB6612FNG Driver: Controls 4x Yellow Gear Motors for mobility.

Servo System: MG90S Servos regulate seed sowing mechanism.

Pumping System: Water pump feeds liquid to the spray nozzle.

Major Features

1. Mini Soil Ploughing

Prepares tilled seedbeds directly in front of the rover during motion.

2. Servo Seed Sowing

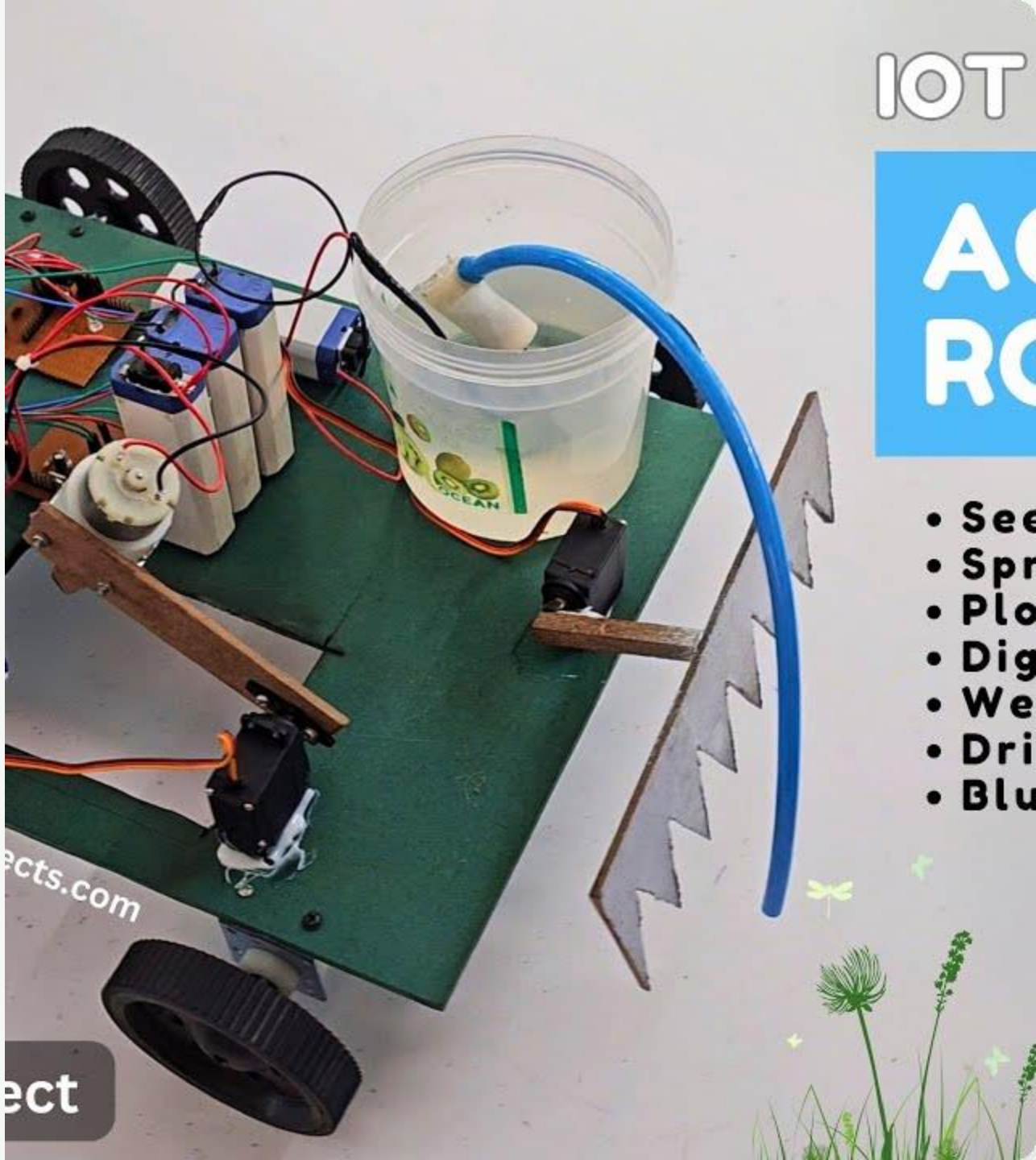
Servo-actuated valve dispenses precise seed counts into tilled furrows.

3. Dual Fluid Spraying

Uniform spraying mechanism handles both irrigation water and pesticides.

4. IoT Mobile Command

Real-time visual navigation and full directional controls (FWD/REV/LR/Stop).



Hardware Components List

Category	Components Details
Processing & Wireless	ESP32 Main Controller, ESP32-CAM Module
Drive & Actuation	4× Yellow Gear Motors + Wheels, TB6612FNG Driver, 2× MG90S Servos
Liquid Mechanism	Mini Water Pump, Water Tank, Spray Pipe & Nozzle
Sensors & Power	Water Sensor, Seed Level Sensor, LDR, MP1584 Buck Converter, 5× 3.7V Li-ion Batteries, 2S & 3S Chargers
Structure & Misc	PVC Whiteboard Chassis, Mini Plough, Veroboard, Power Switch, Buzzer, Jumper Wires

3-Week Development Plan

Timeline	Core Development Focus	Key Milestones & Deliverables
Week 1	Chassis & Base Motion	Assemble PVC chassis, gear motors, TB6612FNG driver, ESP32 integration, battery power setup, and basic manual motion test.
Week 2	Mechanisms & Sensors	Attach mini plough, servo seed dispenser, seed level sensor, water tank, mini pump, and liquid level sensor.
Week 3	IoT, Vision & Field Demo	ESP32-CAM video integration, mobile IoT interface development, dual spraying integration, debugging, and final prototype demonstration.

Estimated Project Budget

Category	Estimated Cost (BDT)
Rover Chassis & Gear Motors	৳1,500
ESP32 + ESP32-CAM Modules	৳1,500
Servos + Motor Driver	৳700
Water Pump + Tank + Accessories	৳500
Sensors (Water/Seed/LDR)	৳300
Batteries & Charging Modules	৳1,200
Electronics, Wiring & Buck Converter	৳1,000
Mechanical Hardware & Plough	৳1,000



Total Estimated Budget

৳7,700 – ৳9,000

Cost-effective multi-functional prototype compared to commercial agricultural machinery.

Expected Outcome

- **Unified Autonomous Rover:** Fully functional mobile agricultural prototype replacing manual labor.
- **Multi-Task Efficiency:** Seamless automated ploughing, seed sowing, and liquid spraying capabilities.
- **Real-Time Monitoring:** Automated alerts for seed availability and liquid reservoir levels.
- **Remote Visual Operation:** Low-latency camera streaming with intuitive mobile IoT remote control.



Future Scope & Enhancements

AI & Autonomy

Implement autonomous path navigation, AI-based weed/plant detection, and obstacle avoidance using ultrasonic sensors.

Smart Analytics

Integrate soil moisture sensors, GPS field mapping, and ML models for crop disease identification.

Cloud Telemetry

Develop a cloud-based IoT dashboard for real-time field data logging, predictive yield estimation, and remote analytics.

Conclusion & Team Members

The Smart Agricultural Rover effectively consolidates essential farming tasks—ploughing, seed sowing, and spraying—into a single IoT-driven mobile system, drastically reducing manual labor while upgrading field monitoring.

Project Team Members

Ahanaf Abtahi 0112420631

Jannatun Nayema Jerin 0112430636

Shawli Sharmin Prionti 0112420327

Nahidul Islam 0112420308

Md. Nurnove 0112420397

Thank you